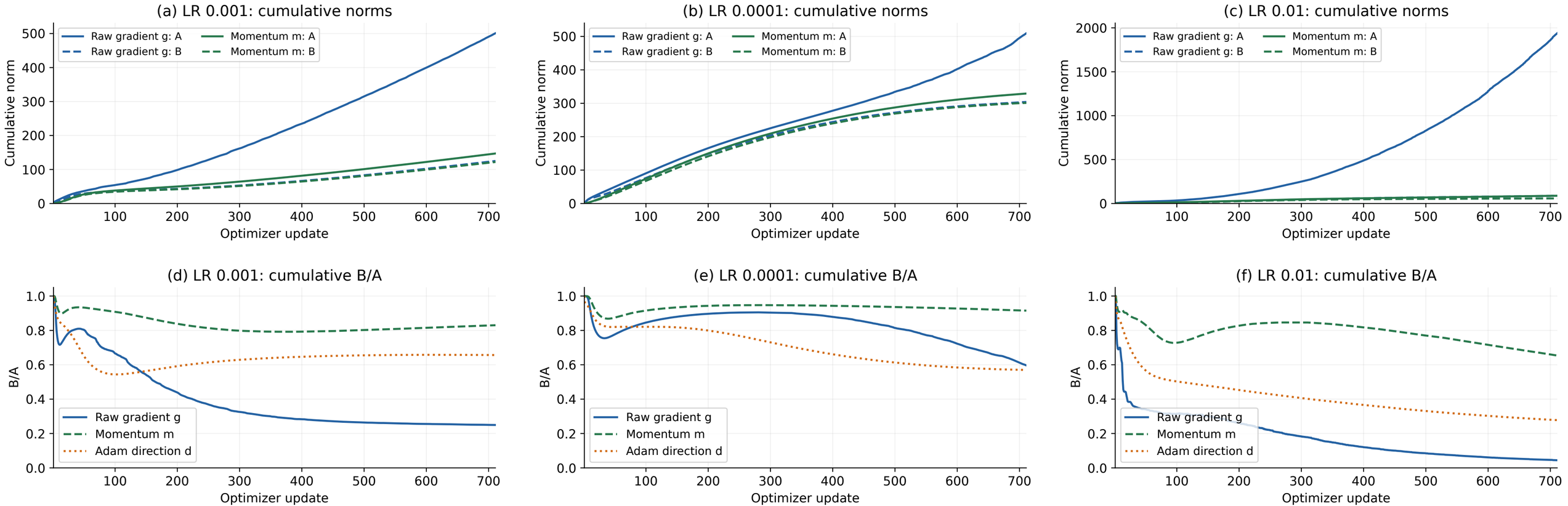
Pythia-14M: geometry within each learning rate
Warmup only; matched initial weights and data order; seed 1234



A = sum of vector norms; B = norm of vector sum. Colors identify raw gradient g, stored momentum m and adaptive direction d.
Cumulative norms: solid A, dashed B; independent linear y-ranges emphasize shape within each LR. B/A: shared linear y-scale.
All curves are unsmoothed. Display: full-sized updates 1-711; partial update 712 retained in results. One matched seed; no cross-seed uncertainty estimated.